SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING COMPUTER SCIENCE AND ENGINEERING LIST OF EXPERIMENTS

SUB CODE & NAME: CSA1712-ARTIFICIAL INTELLIGENCE

2.Write the python program to solve 8-Queen problem

Code:

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```
N = 8
board = [-1] * N

def safe(row, col):

    for i in range(row):

        if board[i] == col:
            return False

        if abs(board[i] - col) == abs(i - row):
            return False

    return True


def solve(row):

    if row == N:
        return True

    for col in range(N):

        if safe(row, col):

            board[row] = col

            if solve(row + 1):
```

```
            board[row] = -1

    return False


solve(0)

print("8-Queen Solution:")

for row in range(N):

    for col in range(N):

        if board[row] == col:
            print("Q", end=" ")
        else:
            print(".", end=" ")

    print()
```

Output:

```
== RESTART: C:/Users/Lekh
8-Queen Solution:
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
```